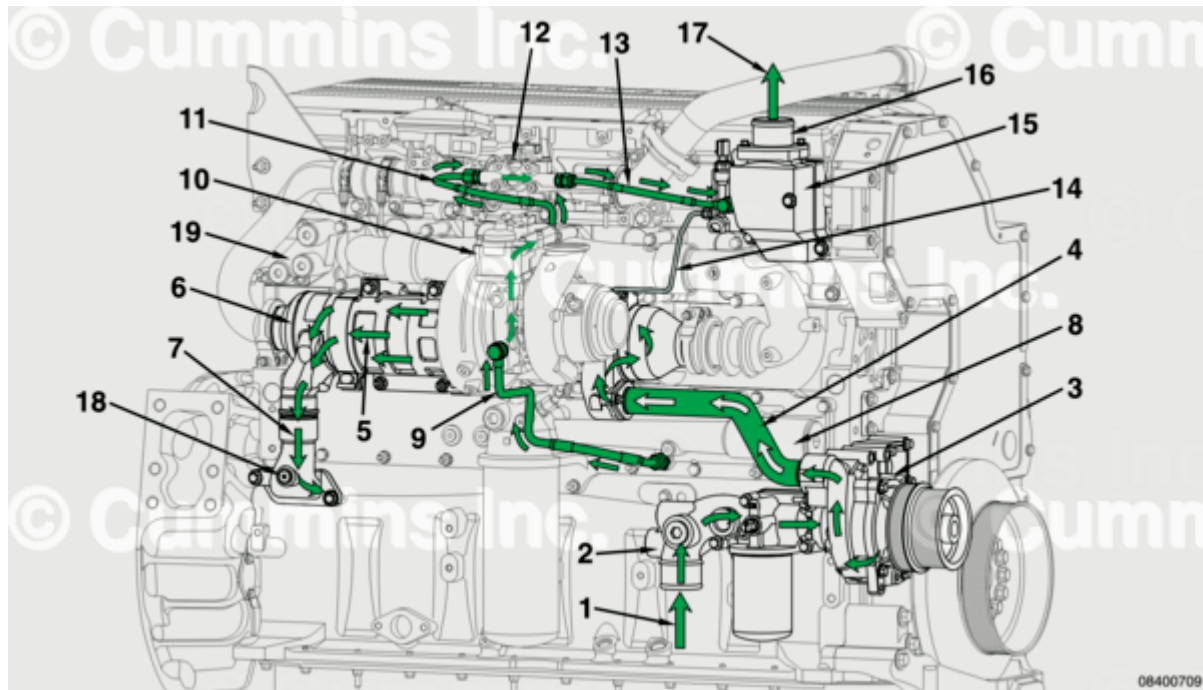
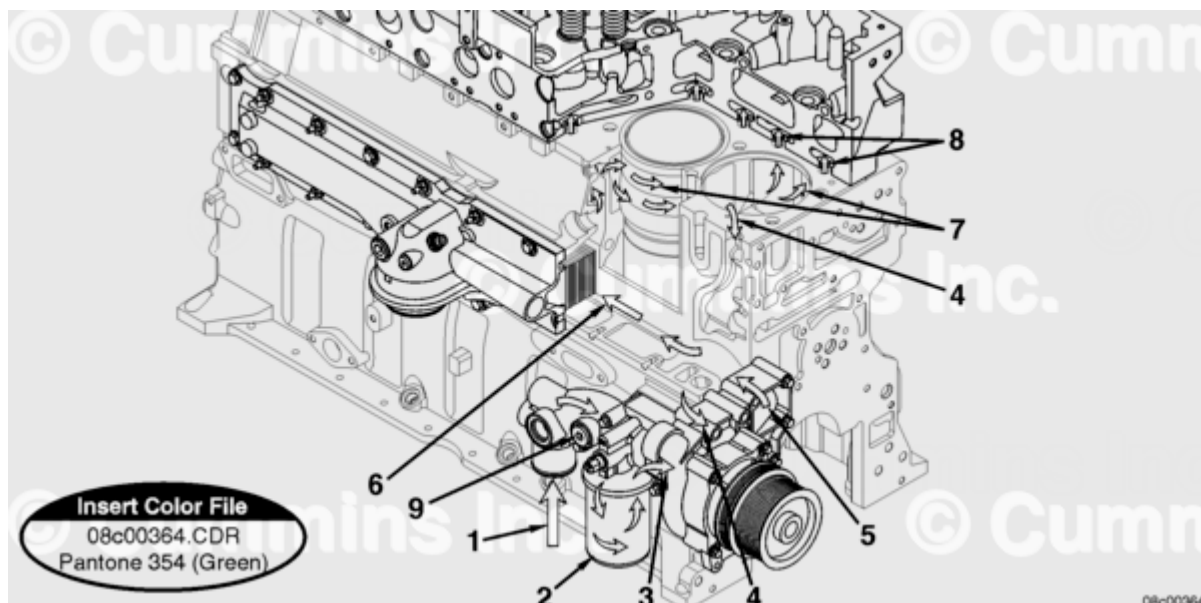


Flow Diagram

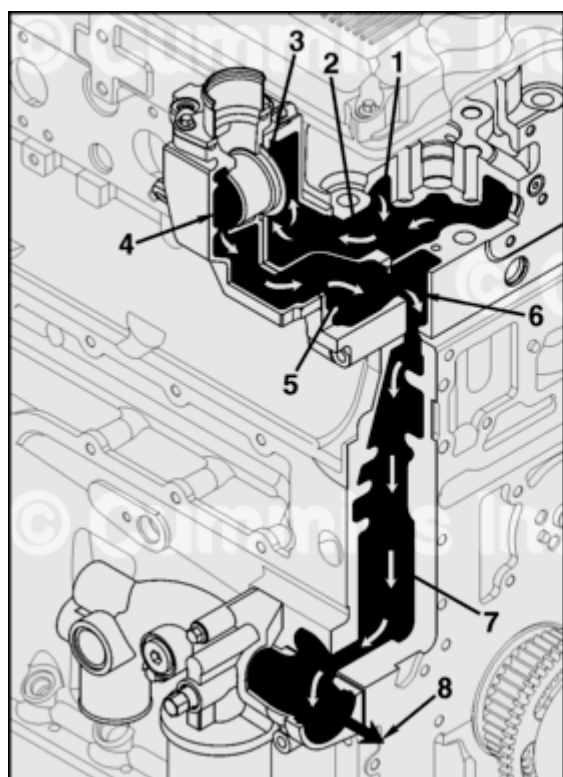


Rear Out Turbocharger

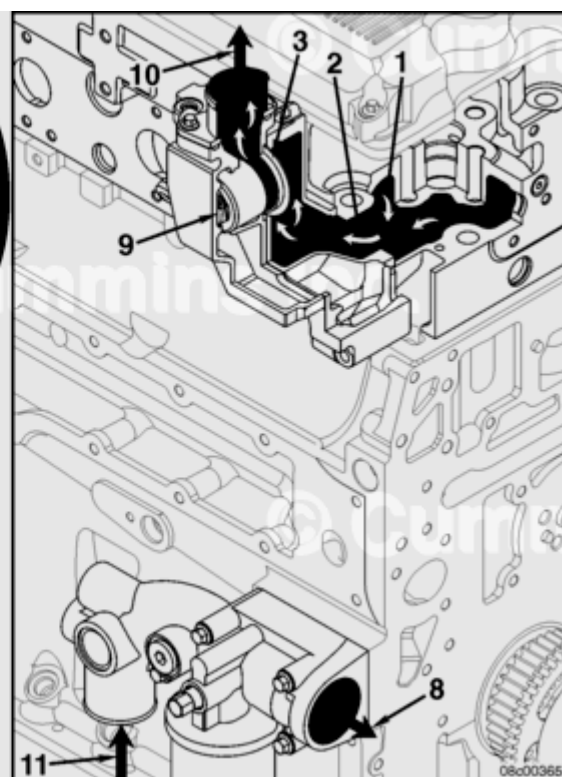
1. Coolant flow from radiator
2. Coolant inlet connection
3. Water pump
4. Coolant supply to exhaust gas recirculation (EGR) cooler
5. Coolant flow through EGR cooler
6. EGR cooler
7. Coolant return to cylinder block
8. Lubricating oil cooler housing
9. Coolant flow to turbocharger bearing housing and turbocharger actuator
10. Turbocharger actuator
11. Coolant flow from turbocharger actuator to EGR mass measurement flow assembly
12. EGR mass measurement flow assembly
13. Coolant flow from EGR mass measurement flow assembly to thermostat housing
14. EGR cooler coolant vent line
15. Thermostat housing
16. Water outlet connection
17. Coolant flow to radiator
18. Selective catalytic reduction (SCR) system coolant supply
19. Original equipment manufacturer (OEM) coolant manifold (supply).



1. Coolant inlet
2. Coolant filter, if equipped
3. Coolant flow to water pump
4. Coolant bypass flow from thermostat
5. Coolant flow from water pump
6. Coolant flow past lubricating oil cooler
7. Coolant flow around cylinder liners
8. Coolant flow to cylinder head
9. OEM coolant manifold and SCR system coolant return.



Insert Color File
08c00365.CDR
Pantone 354 (Green)



1. Coolant flow through cylinder head
2. Coolant flow from cylinder head to thermostat housing
3. Thermostat
4. Bypass open
5. Coolant flow through thermostat housing to bypass passage
6. Coolant bypass flow through cylinder head
7. Coolant bypass passage
8. Coolant bypass flow to water pump
9. Bypass closed
10. Coolant flow to radiator
11. Coolant flow from radiator.